

PHYS 143 – Mechanics and Thermal Physics

The First Law of Thermodynamics

How does a system exchange energy as heat and as work, and how does the first law connect these to its internal energy?

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By the end of this week, you should be able to:

1. Describe a thermodynamic system, state and process, and calculate the work done by a gas from a pV diagram, including its sign and its dependence on the path.
2. Apply the first law, $\Delta U = Q - W$, to a process and to a cycle, using an energy table of Q , W and ΔU .
3. Calculate Q , W and ΔU for the isochoric, isobaric, isothermal and adiabatic processes of an ideal gas, using C_V , C_p and γ .
4. Analyse a multi-step process or cycle in an unfamiliar problem, choosing the physics without being told.

The detailed outcomes are in the checklist at the end of the lecture. **Reading:** Young and Freedman, Ch. 19 (§19.1 to §19.8) and §18.4.

Three contacts:

1 REPRESENT

How does a gas exchange energy with its surroundings?

2 QUANTIFY

What does the first law say, and what does it give for each process?

3 EXTEND AND APPLY

Why do C_p and C_V differ, and how do we treat adiabatic processes and cycles?

Practice: check $\rightarrow$ example $\rightarrow$ independent
 $\rightarrow$ application

Where we start: a five-minute check

Not marked. Answer individually, then we review.

1. A constant force of 20 N moves an object 0.30 m in the direction of the force. Find the work done by the force.
2. Find the area under a horizontal line at $y = 3.0$ from $x = 1.0$ to $x = 5.0$, and under a straight line from $(1.0, 3.0)$ to $(5.0, 1.0)$.
3. Evaluate $\int_{2.0}^{6.0} \frac{dV}{V}$.
4. One mole of a monatomic ideal gas is heated from 300 K to 400 K. Find the change in its total translational kinetic energy.
5. A gas at 1.0×10^5 Pa has its volume doubled at constant temperature. What is its new pressure?
6. Convert 4.0 L to cubic metres, and find $p \Delta V$ in joules for $p = 2.0 \times 10^5$ Pa and $\Delta V = 4.0$ L.

1. $W = Fd = (20)(0.30) = 6.0 \text{ J}$.
2. Horizontal: $3.0 \times 4.0 = 12$. Straight line: average height 2.0, so $2.0 \times 4.0 = 8.0$.
3. $\ln(6.0/2.0) = \ln 3 = 1.10$.
4. $\Delta K_{\text{tot}} = \frac{3}{2}nR\Delta T = \frac{3}{2}(1)(8.314)(100) = 1.25 \times 10^3 \text{ J}$ (Lecture 10).
5. p halves: $0.50 \times 10^5 \text{ Pa}$ (Boyle).
6. $4.0 \times 10^{-3} \text{ m}^3$;
 $p\Delta V = (2.0 \times 10^5)(4.0 \times 10^{-3}) = 800 \text{ J}$.

Work and energy (Lectures 4 and 5), the gas model with $K_{\text{tot}} = \frac{3}{2}nRT$ (Lecture 10) and $Q = mc\Delta T$ (Lecture 9) are reused. Lecture 10 ended with a question: how much work does a gas do when its volume changes, and where does the energy come from? That is the subject of this lecture.

Contact 1: Represent

How does a gas exchange energy with its surroundings?

An opening puzzle: two ways to warm a gas

Poll

The air in a closed bicycle pump can be warmed in two ways: hold the barrel over a flame, or push the handle in quickly. Suppose both end at the *same* pressure, volume and temperature. **Can anyone, by measuring the final state, say which method was used?**

Predict first

Sketch the pump twice, once with a flame under it and once with a hand on the handle. What enters the air in each case? What do p , V and T record?

An opening puzzle: two ways to warm a gas

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Predict first

Sketch the pump twice, once with a flame under it and once with a hand on the handle. What enters the air in each case? What do p , V and T record?

No. The state (p, V, T) records *where the gas is*, not *how it got there*. In the first case energy crossed the wall as **heat**; in the second as **work**. Heat and work describe a *process*; they are not properties of the gas. Today we count both, and relate them to the energy the gas holds.

Defining the vocabulary of thermodynamic work

Quantity	What it is	What it measures	SI unit	Representation
Work done by a gas W	the energy the gas transfers to its surroundings by changing its volume	how much energy the gas gives to, or takes from, the surroundings	J	area under the path on a pV diagram
Path	the sequence of states between the initial and final states	how the change is carried out	—	a curve on the pV diagram
Quasi-static process	a process slow enough that the gas is in equilibrium at every step	whether the path can be drawn at all	—	a continuous line

Interpret: the work done by a gas answers *how much energy did the gas give to the surroundings by pushing the boundary?*

Systems, boundaries and surroundings

- The **system** is the part of the universe we choose to study (the gas in the cylinder). Everything else is the **surroundings**; the **boundary** separates them.
- **Open** system: matter and energy cross the boundary (a cup of hot tea).
- **Closed** system: energy crosses, matter does not (a sealed bottle; the gas in a pump).
- **Isolated** system: neither crosses (an ideal vacuum flask).

Walls (Lecture 9)

A **diathermal** wall lets heat through; an **adiabatic** wall does not. A piston lets the system exchange **work** with the surroundings.

Choose the system first

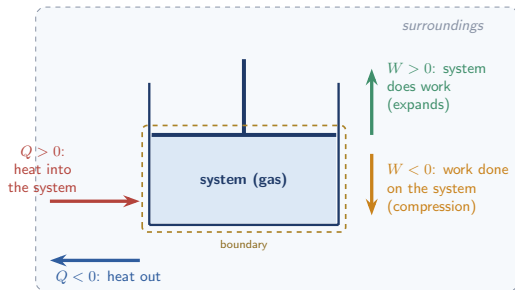
Every energy count depends on where you draw the boundary. Name the system before you write any equation.

- A **state** is specified by the state variables p , V , T , n (Lecture 10). It is defined only when the system is in **equilibrium**: thermal (one temperature), mechanical (uniform pressure) and chemical (no reactions).
- A **process** takes the system from an initial state to a final state. The intermediate states form a **path**.
- A **quasi-static** process is so slow that the gas is in equilibrium at every step. Only then can the path be drawn as a line on a pV diagram.

Physical meaning

An explosion or a sudden rush of gas has no single p and T in the middle of the process, so it has no line on a pV diagram. In this lecture every path is quasi-static, unless we say otherwise.

Energy crossing the boundary: signs for Q and W



Rule: sign convention

$Q > 0$: heat flows **into** the system.

$W > 0$: the system does work **on** its surroundings (it expands).

Negative values mean the reverse.

Caution: not the mechanics sign for work

In mechanics W was the work done *on* a body. Here W is the work done *by* the system. They differ in sign.

TOOLKIT 1 – System and sign convention

- 1. SYSTEM:** name it (the gas), draw the boundary, and say whether it is closed (fixed amount of gas).
 - 2. ENERGY IN OR OUT?** For heat: does energy enter ($Q > 0$) or leave ($Q < 0$)? For work: does the gas push the surroundings ($W > 0$, expansion) or do they push the gas ($W < 0$, compression)?
 - 3. UNITS:** Q , W and ΔU in joules; $1 \text{ Pa m}^3 = 1 \text{ J}$; $1 \text{ cal} = 4.186 \text{ J}$; T in kelvin.
- WATCH OUT:** the mechanics sign for W ; using a temperature interval ΔT where T is needed (and the reverse).
- CHECK:** expansion gives $W > 0$, compression gives $W < 0$; heating the gas gives $Q > 0$.

Concept check: work by a gas before the integral

Decide first

1. Does a gas do work when it expands into a vacuum?
2. Does a gas do work if its volume stays fixed, however high the pressure?
3. Can a gas do work while its temperature stays constant?
4. Is the work done by the gas the same as the work done on the gas?

Concept check: work by a gas before the integral

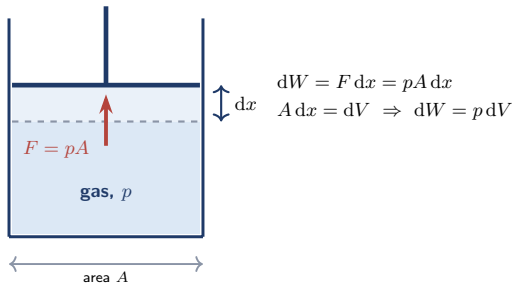
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2. Does a gas do work if its volume stays fixed, however high the pressure?
3. Can a gas do work while its temperature stays constant?
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Answers

1. **No.** There is nothing to push: $p_{\text{ext}} = 0$.
2. **No.** No volume change, no work, whatever p is.
3. **Yes.** An isothermal expansion: heat flows in to pay for it.
4. **They have opposite signs:** $W_{\text{by}} = -W_{\text{on}}$.

Work done by a gas in a volume change



Rule: work done by a gas

$$W = \int_{V_1}^{V_2} p dV$$

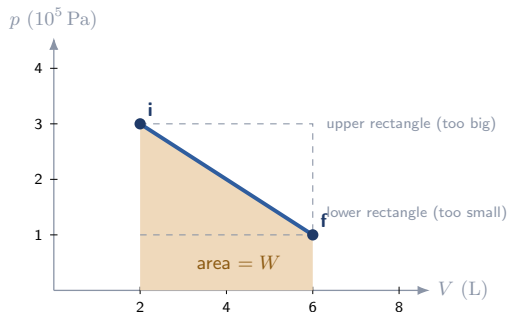
Constant pressure: $W = p(V_2 - V_1)$.

Constant volume: $W = 0$.

Physical meaning

The force on the piston is $F = pA$ (pressure times area). Moving the piston by dx gives $dW = F dx = p dV$. Expansion ($dV > 0$) gives $W > 0$; compression gives $W < 0$. No volume change, no work.

Work is an area on the pV diagram



Rule: work as an area

The work done by the gas is the area under the path on a pV diagram.

Moving to larger V gives $W > 0$;
moving to smaller V gives $W < 0$.

Common error

Using $p \Delta V$ with the final (or the initial) pressure on a sloping path. p changes along the path.

EQUATION – work by a gas

$$W = \int_{V_1}^{V_2} p \, dV$$

Meaning. the energy the gas transfers to the surroundings as it pushes the boundary.

Variables. $p(V)$ pressure along the path; V_1 , V_2 the end volumes.

Units. $\text{Pa m}^3 = \text{J}$.

Interpretation. the area under the path; positive on expansion, negative on compression, zero at constant volume; it depends on the path.

Example. the straight-line path: 800 J.

Predict first

A gas expands from 2.0 L to 6.0 L. Predict which gives the larger work: staying at the high initial pressure, or at the low final pressure.

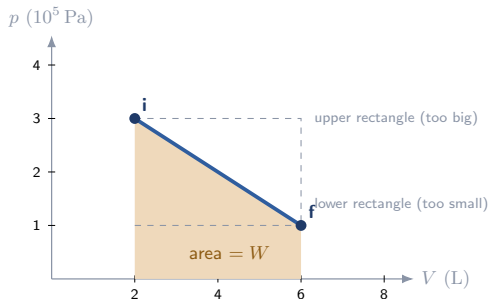
TOOLKIT 2 – Work from a pV diagram

1. **SKETCH** the path with an arrow from the initial to the final state; mark the volumes.
2. **SPLIT INTO LEGS:** horizontal $W = p\Delta V$; vertical $W = 0$; straight sloping line: average pressure times ΔV (a trapezoid); curve: integrate (isothermal: next frames).
3. **SIGN:** moving right (V increases) is positive; moving left is negative.
4. **UNITS:** p in pascals and V in m^3 (litres must be converted).
5. **CYCLE:** the net work is the enclosed area; clockwise is positive, counter-clockwise is negative.

WATCH OUT: V left in litres; forgetting the sign of a leg that goes to smaller V .

CHECK: the answer lies between the areas of the rectangles that bound the path.

Example 1: a straight-line path



A gas expands from an initial state i ($V = 2.0 \text{ L}$, $p = 3.0 \times 10^5 \text{ Pa}$) to a final state f ($V = 6.0 \text{ L}$, $p = 1.0 \times 10^5 \text{ Pa}$) along a straight line on the pV diagram. Find the work done by the gas.

Predict first

Is W positive or negative? It must lie between the two rectangles shown. Between which two values in joules?

Set up: $\Delta V = (6.0 - 2.0) \times 10^{-3} = 4.0 \times 10^{-3} \text{ m}^3$. The bounding rectangles:
 $p_f \Delta V = (1.0 \times 10^5)(4.0 \times 10^{-3}) = 400 \text{ J}$ and $p_i \Delta V = 1200 \text{ J}$.

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Execute: the area is a trapezoid, with average height $\frac{1}{2}(3.0 + 1.0) \times 10^5 = 2.0 \times 10^5 \text{ Pa}$:

$$W = p_{\text{av}} \Delta V = (2.0 \times 10^5)(4.0 \times 10^{-3}) = 800 \text{ J}.$$

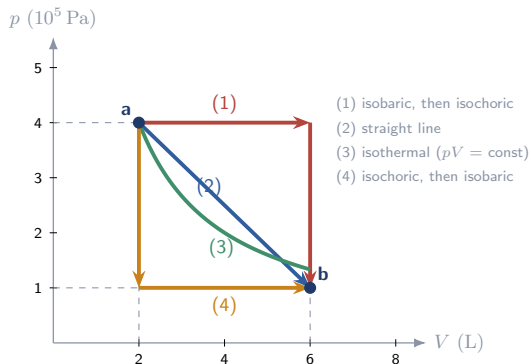
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$$W = p_{\text{av}} \Delta V = (2.0 \times 10^5)(4.0 \times 10^{-3}) = 800 \text{ J}.$$

Evaluate: $W > 0$ for an expansion, and $400 < 800 < 1200 \text{ J}$, exactly halfway between the rectangles because the path is straight. The units work: $\text{Pa m}^3 = \text{J}$. ✓

The work depends on the path



A gas goes from a (2.0 L, 4.0×10^5 Pa) to b (6.0 L, 1.0×10^5 Pa) by four different paths. Same start, same end.

Predict first

Which path gives the greatest work, and which the least? Is the work the same for all four?

Three of the four paths

$\Delta V = 4.0 \times 10^{-3} \text{ m}^3$ on every path.

Path (1) (isobaric at $4.0 \times 10^5 \text{ Pa}$, then vertical):

$$W_1 = p_a \Delta V = (4.0 \times 10^5)(4.0 \times 10^{-3}) = \mathbf{1600 \text{ J}}.$$

Path (2) (straight line): $W_2 = \frac{1}{2}(4.0 + 1.0) \times 10^5 \times 4.0 \times 10^{-3} = \mathbf{1000 \text{ J}}.$

Path (4) (vertical drop at 2.0 L , then isobaric at $1.0 \times 10^5 \text{ Pa}$): $W_4 = p_b \Delta V = \mathbf{400 \text{ J}}.$

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Path (4) (vertical drop at 2.0 L , then isobaric at $1.0 \times 10^5 \text{ Pa}$): $W_4 = p_b \Delta V = \mathbf{400 \text{ J}}.$

Path (3) is a curve. Its area needs an integral, which we do next. For the ideal gas the curve is an isotherm, because $p_a V_a = p_b V_b = 800 \text{ J}$.

Preview

Three different works already: the work is *not* fixed by the end states.

Rule: work in an isothermal process

$$W = \int_{V_1}^{V_2} \frac{nRT}{V} dV = nRT \ln \frac{V_2}{V_1} = p_1 V_1 \ln \frac{V_2}{V_1}$$

Ideal gas at constant T : $p = nRT/V$, with T taken out of the integral.

Path (3): $p_a V_a = 800 \text{ J}$, $V_2/V_1 = 3$:

$$W_3 = 800 \ln 3 = 800(1.0986) = \mathbf{879 \text{ J}}.$$

Physical meaning

Expansion ($V_2 > V_1$) gives $\ln > 0$, so $W > 0$; compression gives $W < 0$.

Constant temperature does *not* mean zero work.

Check: 879 J lies between 1000 J (straight line, above) and 400 J (lowest path), as the sketch requires.

EQUATION – isothermal work, ideal gas

$$W = nRT \ln \frac{V_2}{V_1} = p_1 V_1 \ln \frac{V_2}{V_1}$$

Meaning. the work as an ideal gas expands or is compressed at constant temperature.

Variables. n moles; T kelvin; V_1 , V_2 end volumes.

Units. J.

Interpretation. positive for expansion, negative for compression; constant T does not mean zero work.

Example. path (3): $800 \ln 3 = 879 \text{ J}$.

Predict first

For the same end volumes, predict whether the isothermal work lies above or below the straight-line path.

Same end points, four different works

Path	Description	W (J)
(1)	isobaric, then isochoric	1600
(2)	straight line	1000
(3)	isothermal	879
(4)	isochoric, then isobaric	400

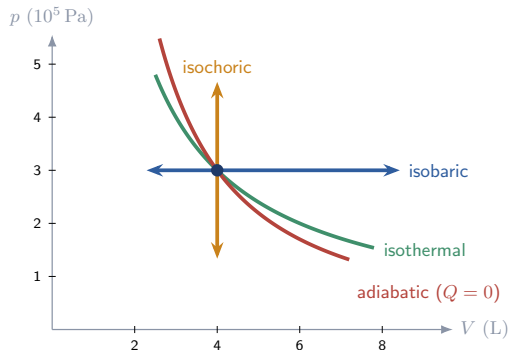
Rule: work is not a state property

The work depends on the path, not only on the end states. We write W , never " ΔW ".

Question for Contact 2

The same is true of heat. So what *is* the same on all four paths? Think about what changes when the gas goes from a to b.

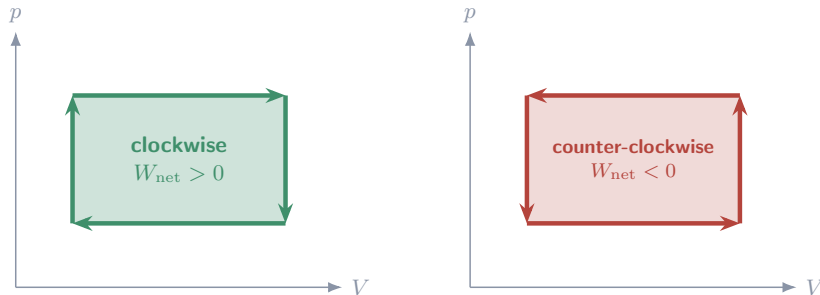
The four special processes on a pV diagram



Process	Fixed	Path
isochoric	V	vertical
isobaric	p	horizontal
isothermal	T	hyperbola
adiabatic	$Q = 0$	steeper curve

The first three are named for the quantity held fixed. The adiabatic curve falls faster than the isotherm (Contact 3).

Cycles: net work is the enclosed area



Rule: work in a cycle

In a cycle the gas returns to its initial state. The net work is the **area enclosed** by the path: positive if the path runs clockwise, negative if counter-clockwise.

Independent problem

A gas goes round the cycle $a \rightarrow b \rightarrow c \rightarrow a$. a : 2.0 L, 3.0×10^5 Pa. $a \rightarrow b$: constant pressure to 5.0 L. $b \rightarrow c$: constant volume to 1.0×10^5 Pa. $c \rightarrow a$: straight line back to a . Find the work for each leg and the net work.

Predict first

Sketch the cycle. Does the path run clockwise or counter-clockwise? What sign must the net work have?

Independent problem

A gas goes round the cycle $a \rightarrow b \rightarrow c \rightarrow a$. a : 2.0 L, 3.0×10^5 Pa. $a \rightarrow b$: constant pressure to 5.0 L. $b \rightarrow c$: constant volume to 1.0×10^5 Pa. $c \rightarrow a$: straight line back to a . Find the work for each leg and the net work.

Predict first

Sketch the cycle. Does the path run clockwise or counter-clockwise? What sign must the net work have?

$a \rightarrow b$: $W = p\Delta V = (3.0 \times 10^5)(3.0 \times 10^{-3}) = 900$ J. $b \rightarrow c$: $W = 0$. $c \rightarrow a$: compression, $W = \frac{1}{2}(1.0 + 3.0) \times 10^5 \times (-3.0 \times 10^{-3}) = -600$ J.

Net: $900 + 0 - 600 = 300$ J. **Evaluate:** clockwise gives $W > 0$; the enclosed triangle has area $\frac{1}{2}(3.0 \times 10^{-3})(2.0 \times 10^5) = 300$ J. ✓

Choose, then explain

A gas expands from V_1 to V_2 along a path on a pV diagram, and is then compressed back to the original state along exactly the same path. What is the net work done by the gas?

- A. positive B. negative C. zero D. it depends on the path

Predict first

Sketch the path with arrows both ways. What happens to the sign of dV on the return trip?

Choose, then explain

A gas expands from V_1 to V_2 along a path on a pV diagram, and is then compressed back to the original state along exactly the same path. What is the net work done by the gas?

- A. positive B. negative C. zero D. it depends on the path

Predict first

Sketch the path with arrows both ways. What happens to the sign of dV on the return trip?

C. On the return trip $dV < 0$ at each pressure the gas had before, so the integral changes sign: $W_{\text{back}} = -W_{\text{forward}}$ and the enclosed area is zero. The result holds for any path, which is why it does not depend on it (D is wrong).

Common mistakes and the correct habit

Mistake

Using the mechanics sign for W (work done on).

Using $p \Delta V$ with the wrong pressure on a sloping path.

Leaving V in litres.

Writing ΔW .

Saying a free expansion does work.

Instead

Here W is the work done *by* the system.

Use the area under the path.

Convert to m^3 : then $\text{Pa m}^3 = \text{J}$.

Work depends on the path; write W .

There is nothing to push: $W = 0$.

Contact 1: what we can now do

System, state, process



Signs: Q into, W by the system



Work: $W = \int p dV = \text{area under the path}$



Depends on the path; cycle: enclosed area

You should be able to

- name the system and classify it as open, closed or isolated;
- apply the sign convention for Q and W ;
- find W from a pV diagram, with sign;
- show that W depends on the path.

Connect: Lecture 12

The area enclosed by a cycle is the net work of an engine (Lecture 12).

Contact 2: Quantify

*What does the first law say, and
what does it give for each process?*

An opening question: what is conserved?

Poll

For the four paths from a to b, the work done by the gas was 1600, 1000, 879 and 400 J. The heat that flowed into the gas was also measured on each path. Do the heat inputs come out the same, or different? And is *anything* the same on all four paths?

Predict first

The gas starts and finishes in the same two states every time. Which quantity can depend only on those two states? Which cannot?

An opening question: what is conserved?

Poll

For the four paths from a to b, the work done by the gas was 1600, 1000, 879 and 400 J. The heat that flowed into the gas was also measured on each path. Do the heat inputs come out the same, or different? And is *anything* the same on all four paths?

Predict first

The gas starts and finishes in the same two states every time. Which quantity can depend only on those two states? Which cannot?

The heat inputs are different, just as the works are. But the *difference* (heat in, minus work out) is the same on every path. It is the change in a property of the gas called the **internal energy**. We name it first, then see the law.

- The **internal energy** U is the sum of all the kinetic and potential energies of the molecules in the system, relative to the centre of mass of the system.
- Kinetic: translation, rotation, vibration.
Potential: forces between molecules (and chemical energy).
- U is a **state function**: it has one value for each equilibrium state, so $\Delta U = U_f - U_i$ does not depend on the path.

Common errors

- (1) “A body contains heat.” It contains *internal energy*; heat and work are energy *in transit*.
- (2) Lifting a glass of water raises its potential energy in the Earth’s field, but not its internal energy.

Decide first

1. Can the temperature of a gas rise without any heat entering it?
2. Can heat flow into a gas without its temperature rising?
3. Does a gas “contain” heat or work?

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Answers

1. **Yes.** Compress it quickly: work is done on it.
2. **Yes.** An isothermal expansion: the heat becomes work.
3. **No.** It contains internal energy. Heat and work are energy in transit.

Rule: the first law

$$\Delta U = Q - W \quad \text{or, for a small step,} \quad dU = dQ - dW$$

The change in internal energy equals the heat added to the system minus the work done by the system.

Physical meaning

This is the conservation of energy (Lecture 5), with heat included. Q and W separately depend on the path; their difference does not.

Event	Sign	ΔU
heat in, no work	$Q > 0$	up
expands, no heat	$W > 0$	down
compressed, no heat	$W < 0$	up
heat out, no work	$Q < 0$	down

EQUATION – first law of thermodynamics

$$\Delta U = Q - W$$

Meaning. conservation of energy: heat added minus work done by the system changes its internal energy.

Variables. $\Delta U = U_f - U_i$; $Q > 0$ into the system; $W > 0$ by the system.

Units. J for each term.

Interpretation. Q and W depend on the path; ΔU does not.

Example. path abd: $4400 - 1800 = 2600$ J.

Predict first

A gas absorbs 500 J of heat and does 200 J of work.

Predict the sign and size of ΔU before calculating.

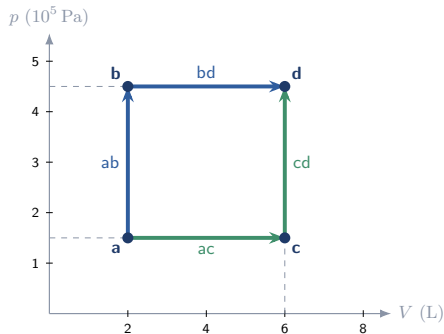
TOOLKIT 3 – The energy ledger

1. **ROWS:** one row for each step of the process (or for each path); columns Q , W , ΔU .
2. **FILL IN** what is known; on every row $\Delta U = Q - W$.
3. **STATE FUNCTION:** ΔU between two states is the same on every path. Find it on a path where it is easy, then use it on the other.
4. **TOTALS:** add each column. The first law holds for the totals; for a cycle, the total ΔU is zero.

WATCH OUT: a wrong sign in W ; adding a row twice; using Q where ΔU is meant.

CHECK: $\sum Q - \sum W = \sum \Delta U$ on the totals line.

Example 2: two paths, one substance



a: 2.0 L, $1.5 \times 10^5 \text{ Pa}$. b: 2.0 L, $4.5 \times 10^5 \text{ Pa}$. d: 6.0 L, $4.5 \times 10^5 \text{ Pa}$. c: 6.0 L, $1.5 \times 10^5 \text{ Pa}$. In process ab , 1800 J of heat is added; in bd , 2600 J. Find (a) ΔU for ab , (b) ΔU for abd , (c) the total heat added along acd . (Any substance: the first law is not limited to ideal gases.)

Predict first

Path acd does less work than path abd . Will it need more heat or less heat than abd to reach the same state d ?

Two paths: the ledger

Path abd

Step	Process	Q	W	ΔU
ab	isochoric	1800	0	1800
bd	isobaric	2600	1800	800
abd	total	4400	1800	2600

$$W_{bd} = p\Delta V = (4.5 \times 10^5)(4.0 \times 10^{-3}) = 1800 \text{ J.}$$

Two paths: the ledger

Path abd

Step	Process	Q	W	ΔU
ab	isochoric	1800	0	1800
bd	isobaric	2600	1800	800
abd	total	4400	1800	2600

$$W_{bd} = p\Delta V = (4.5 \times 10^5)(4.0 \times 10^{-3}) = 1800 \text{ J.}$$

Path acd

Step	Process	Q	W	ΔU
ac	isobaric		600	
cd	isochoric		0	
acd	total	3200	600	2600

$$W_{ac} = (1.5 \times 10^5)(4.0 \times 10^{-3}) = 600 \text{ J;}$$

$$\Delta U_{acd} = \Delta U_{abd}, \text{ so } Q_{acd} = 2600 + 600.$$

Two paths: the ledger

Path abd

Step	Process	Q	W	ΔU
ab	isochoric	1800	0	1800
bd	isobaric	2600	1800	800
abd	total	4400	1800	2600

$$W_{bd} = p\Delta V = (4.5 \times 10^5)(4.0 \times 10^{-3}) = 1800 \text{ J.}$$

Path acd

Step	Process	Q	W	ΔU
ac	isobaric		600	
cd	isochoric		0	
acd	total	3200	600	2600

$$W_{ac} = (1.5 \times 10^5)(4.0 \times 10^{-3}) = 600 \text{ J;}$$

$$\Delta U_{acd} = \Delta U_{abd}, \text{ so } Q_{acd} = 2600 + 600.$$

Evaluate: (a) 1800 J; (b) 2600 J; (c) 3200 J. Less work means less heat is needed to reach the same state, as predicted ($3200 < 4400$ J). The empty cells could be filled only if we knew more about the substance. ✓

Rule: the first law for a cycle

$$\Delta U = 0 \quad \Rightarrow \quad Q_{\text{net}} = W_{\text{net}} = \text{area enclosed}$$

Clockwise: $W_{\text{net}} > 0$ and heat is converted to work overall. Counter-clockwise: $W_{\text{net}} < 0$ and work is converted to heat leaving the system.

Physical meaning

After a full cycle the system is unchanged, so nothing is stored. Whatever net work it does has been paid for by net heat taken in.

Common error

“In a cycle $Q = 0$ or $W = 0$.”
Neither is zero. It is ΔU that is zero, so $Q_{\text{net}} = W_{\text{net}}$.

Your turn: a counter-clockwise cycle

Independent problem

A gas goes round a counter-clockwise cycle on a pV diagram. The enclosed area is 350 J. (a) Why is the net work negative? (b) Find ΔU , W_{net} and Q_{net} . (c) Does heat enter or leave the gas overall?

Predict first

Sketch the loop with its arrows. Is the surroundings doing work on the gas, or the gas on the surroundings?

Your turn: a counter-clockwise cycle

Independent problem

A gas goes round a counter-clockwise cycle on a pV diagram. The enclosed area is 350 J. (a) Why is the net work negative? (b) Find ΔU , W_{net} and Q_{net} . (c) Does heat enter or leave the gas overall?

Predict first

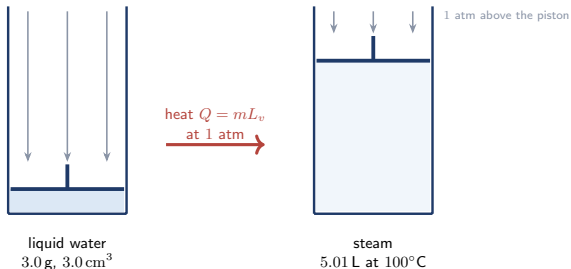
Sketch the loop with its arrows. Is the surroundings doing work on the gas, or the gas on the surroundings?

(a) Counter-clockwise: the compression legs lie above the expansion legs, so more work is done on the gas than by it.

(b) $\Delta U = 0$ (cycle), $W_{\text{net}} = -350 \text{ J}$, so $Q_{\text{net}} = \Delta U + W_{\text{net}} = -350 \text{ J}$.

(c) **Evaluate:** $Q_{\text{net}} < 0$: heat leaves the gas overall, equal to the work done on it. A refrigerator runs on a cycle of this kind (Lecture 12). ✓

The first law and a change of phase



3.0 g of water (3.0 cm³) is boiled at 1.00 atm into steam of volume 5.01 L (a volume of 1671 cm³ per gram). The latent heat is $L_v = 2.256 \times 10^6$ J/kg. Find (a) the heat added, (b) the work done by the water, (c) the change in internal energy.

Predict first

The temperature stays at 100°C. Is most of the heat used to push back the atmosphere, or to change U ?

$$(a) \quad Q = mL_v = (3.0 \times 10^{-3})(2.256 \times 10^6) = 6.77 \times 10^3 \text{ J}.$$

(a) $Q = mL_v = (3.0 \times 10^{-3})(2.256 \times 10^6) = 6.77 \times 10^3 \text{ J}.$

(b) constant pressure, $p = 1.013 \times 10^5 \text{ Pa}:$

$$W = p(V_2 - V_1) = (1.013 \times 10^5)(5.01 \times 10^{-3} - 3 \times 10^{-6}) = 5.1 \times 10^2 \text{ J}.$$

(a) $Q = mL_v = (3.0 \times 10^{-3})(2.256 \times 10^6) = 6.77 \times 10^3 \text{ J}.$

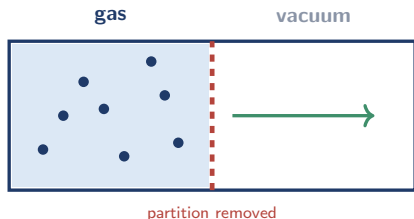
(b) constant pressure, $p = 1.013 \times 10^5 \text{ Pa}:$

$$W = p(V_2 - V_1) = (1.013 \times 10^5)(5.01 \times 10^{-3} - 3 \times 10^{-6}) = 5.1 \times 10^2 \text{ J}.$$

(c) $\Delta U = Q - W = 6768 - 507 = 6.26 \times 10^3 \text{ J}.$

Evaluate: only 7.5% of the heat is the work against the atmosphere; 92.5% goes into U . The molecular kinetic energy does not change (the temperature is constant), so the 6.26 kJ raises the *potential* energy of the molecules as they are pulled apart. Heat was added and the temperature did not rise. ✓

Internal energy of an ideal gas



In a **free expansion** the gas rushes into a vacuum: $W = 0$ (nothing to push), and with insulated walls $Q = 0$, so $\Delta U = 0$. Experiment: the temperature of a dilute gas does not change.

Rule: internal energy of an ideal gas

U depends on T only, not on p or V . For a monatomic ideal gas (no forces, so no potential energy; Lecture 10):

$$U = \frac{3}{2}nRT \quad \Delta U = \frac{3}{2}nR\Delta T \quad \text{in every process.}$$

EQUATION – internal energy, ideal gas

$$U = \frac{3}{2}nRT, \quad \Delta U = nC_V\Delta T \quad (\text{every process})$$

Meaning. the energy stored in the molecular motion of the gas.

Variables. n moles; T kelvin; C_V molar heat capacity at constant volume.

Units. J.

Interpretation. depends on T only; $\Delta U = 0$ on an isotherm.

Example. leg BC: $\Delta U = -2250 \text{ J}$.

Predict first

A gas is compressed adiabatically. Predict the sign of ΔU and of ΔT .

Back to the four paths

States a and b lie on the same isotherm ($p_a V_a = p_b V_b = 800 \text{ J}$), so for an ideal gas $T_a = T_b$ and $\Delta U = 0$. The first law then gives $Q = W$ on every path:

Path	$W \text{ (J)}$	$\Delta U \text{ (J)}$	$Q \text{ (J)}$
(1)	1600	0	1600
(2)	1000	0	1000
(3)	879	0	879
(4)	400	0	400

Physical meaning

W and Q change from path to path, in step. $\Delta U = Q - W = 0$ does not. The gas must take in heat on every path, because it does work and returns to the same internal energy.

Common error

“Isothermal means no heat flow.”
Here $Q = W > 0$: heat flows in to pay for the work.

The four processes of an ideal gas

Rule: the first law for each process (monatomic gas, $\Delta U = \frac{3}{2}nR\Delta T$)

Process	Condition	W	ΔU	Q
isochoric	V constant	0	$\frac{3}{2}nR\Delta T$	$= \Delta U$
isobaric	p constant	$p\Delta V = nR\Delta T$	$\frac{3}{2}nR\Delta T$	$\frac{5}{2}nR\Delta T$
isothermal	T constant	$nRT \ln \frac{V_2}{V_1}$	0	$= W$
adiabatic	$Q = 0$	$= -\Delta U$	$\frac{3}{2}nR\Delta T$	0

Isobaric: $Q = \Delta U + W = \frac{3}{2}nR\Delta T + nR\Delta T$. In the adiabatic process the work comes out of U : the gas cools when it expands and warms when it is compressed (calculation in Contact 3).

TOOLKIT 4 – Recognising the process

1. READ THE WORDS: “rigid container” means V constant; “free piston under the atmosphere” or “heated in the open” means p constant; “slowly, in a thermal bath” means T constant; “insulated” or “very fast” means $Q \approx 0$.

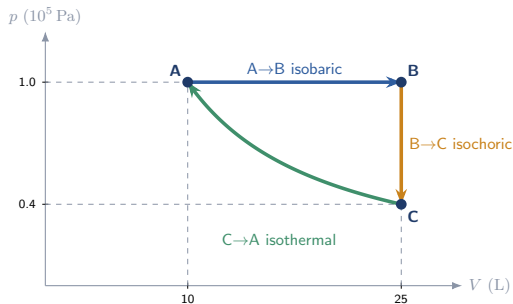
2. READ THE DIAGRAM: vertical (isochoric), horizontal (isobaric), hyperbola $pV = \text{const}$ (isothermal), a curve steeper than the hyperbola (adiabatic).

3. WRITE THE CONSEQUENCE from the table: which of W , ΔU , Q is zero or known at once?

WATCH OUT: isothermal ($\Delta T = 0$) confused with adiabatic ($Q = 0$); using $\Delta U = 0$ for any process except an isotherm of an ideal gas.

CHECK: for an ideal gas, the sign of ΔU matches the sign of ΔT .

Example 3: a three-process cycle



0.40 mol of helium (a monatomic ideal gas) starts at A (1.0×10^5 Pa, 10.0 L). A $\rightarrow$ B: isobaric expansion to 25.0 L. B $\rightarrow$ C: isochoric cooling until the temperature equals that at A. C $\rightarrow$ A: isothermal compression back to A. Find Q , W and ΔU for each leg.

Predict first

Sketch the cycle. Which legs absorb heat and which reject it? Is the net work positive?

Three-process cycle: the ledger

Set up: $p_A V_A = 1000 \text{ J}$, $T_A = \frac{1000}{(0.40)(8.314)} = 300.7 \text{ K}$;

$$T_B = \frac{(1.0 \times 10^5)(25.0 \times 10^{-3})}{(0.40)(8.314)} = 751.7 \text{ K}; T_C = T_A \text{ so } p_C = \frac{p_A V_A}{V_C} = 0.40 \times 10^5 \text{ Pa}.$$

Leg	Process	$Q \text{ (J)}$	$W \text{ (J)}$	$\Delta U \text{ (J)}$
A→B	isobaric	3750	1500	2250
B→C	isochoric	-2250	0	-2250
C→A	isothermal	-916	-916	0
cycle	total	584	584	0

Three-process cycle: the ledger

Set up: $p_A V_A = 1000 \text{ J}$, $T_A = \frac{1000}{(0.40)(8.314)} = 300.7 \text{ K}$;

$$T_B = \frac{(1.0 \times 10^5)(25.0 \times 10^{-3})}{(0.40)(8.314)} = 751.7 \text{ K}; T_C = T_A \text{ so } p_C = \frac{p_A V_A}{V_C} = 0.40 \times 10^5 \text{ Pa}.$$

Leg	Process	$Q \text{ (J)}$	$W \text{ (J)}$	$\Delta U \text{ (J)}$
A→B	isobaric	3750	1500	2250
B→C	isochoric	-2250	0	-2250
C→A	isothermal	-916	-916	0
cycle	total	584	584	0

AB: $W = p\Delta V = 1500$, $\Delta U = \frac{3}{2}(p_B V_B - p_A V_A) = \frac{3}{2}(2500 - 1000) = 2250$. BC: $\Delta U = \frac{3}{2}(1000 - 2500)$. CA: $W = 1000 \ln(10/25) = -916$.

Evaluate: the cycle closes: $\sum \Delta U = 0$ and $\sum Q = \sum W = 584 \text{ J}$, which is the enclosed area. Heat is absorbed only in AB. ✓

Your turn: an isothermal compression

Independent problem

0.25 mol of an ideal gas at 350 K is compressed isothermally and slowly from 12 L to 4.0 L. Find the initial and final pressures, and W , ΔU and Q .

Predict first

Sketch the path on a pV diagram. By what factor does the pressure change? Is W positive or negative? Does heat enter or leave the gas?

Your turn: an isothermal compression

Independent problem

0.25 mol of an ideal gas at 350 K is compressed isothermally and slowly from 12 L to 4.0 L. Find the initial and final pressures, and W , ΔU and Q .

Predict first

Sketch the path on a pV diagram. By what factor does the pressure change? Is W positive or negative? Does heat enter or leave the gas?

$$p_1 = \frac{nRT}{V_1} = \frac{(0.25)(8.314)(350)}{12 \times 10^{-3}} = 6.06 \times 10^4 \text{ Pa}; \quad p_2 = 3p_1 = 1.82 \times 10^5 \text{ Pa}.$$

$$W = nRT \ln \frac{V_2}{V_1} = (727.5) \ln \frac{4.0}{12} = -799 \text{ J}; \quad \Delta U = 0; \quad Q = W = -799 \text{ J}.$$

Evaluate: work is done *on* the gas ($W < 0$) and the same amount of energy leaves as heat ($Q < 0$), so T stays constant. ✓

Choose, then explain

An ideal gas is compressed to half its volume, once slowly in a large thermal bath and once very quickly in an insulated cylinder. Which statement is correct?

- A. Isothermal: T rises. Adiabatic: T constant.
- B. Isothermal: T constant and heat leaves the gas. Adiabatic: T rises and no heat flows.
- C. Both: T constant.
- D. Isothermal: no heat flows. Adiabatic: heat leaves the gas.
- E. Both: T rises.

Predict first

Work is done on the gas in both. In which case can the energy leave as heat?

Choose, then explain

An ideal gas is compressed to half its volume, once slowly in a large thermal bath and once very quickly in an insulated cylinder. Which statement is correct?

- A. Isothermal: T rises. Adiabatic: T constant.
- B. Isothermal: T constant and heat leaves the gas. Adiabatic: T rises and no heat flows.
- C. Both: T constant.
- D. Isothermal: no heat flows. Adiabatic: heat leaves the gas.
- E. Both: T rises.

Predict first

Work is done on the gas in both. In which case can the energy leave as heat?

B. Isothermal: $\Delta U = 0$, so $Q = W < 0$. Adiabatic: $Q = 0$, so $\Delta U = -W > 0$ and T rises.
Answer D swaps the two.

Common mistakes and the correct habit

Mistake

Saying a gas “contains” heat or work.

Saying $Q = 0$ or $W = 0$ in a cycle.

Saying isothermal means no heat flow.

Using $\Delta U = 0$ for any process.

Taking $\Delta U = Q$ in every process.

Instead

It contains internal energy; heat and work are in transit.

$\Delta U = 0$, so $Q_{\text{net}} = W_{\text{net}}$.

T constant: $Q = W$ for an ideal gas.

Only on an isotherm of an ideal gas.

$\Delta U = Q - W$; $\Delta U = Q$ only at constant volume.

Contact 2: what we can now do

Internal energy U : state function

⇓

First law: $\Delta U = Q - W$; the ledger

⇓

Cycle: $\Delta U = 0$, $Q_{\text{net}} = W_{\text{net}}$

⇓

Ideal gas: $U(T)$; four processes

You should be able to

- say what U is, and what heat and work are not;
- apply $\Delta U = Q - W$; build the ledger;
- treat a cycle and the four processes.

Connect: Lectures 5 and 12

$\Delta U = Q - W$ extends energy conservation (Lecture 5); the ledger returns for heat engines and refrigerators (Lecture 12).

Contact 3: Extend and apply

Why do heat capacities differ, what is an adiabatic process, and how do we analyse a cycle?

An opening question: heat per kelvin

Poll

In the three-process cycle of Contact 2, the isobaric heating $A \rightarrow B$ took 3750 J and the isochoric cooling $B \rightarrow C$ released 2250 J. The temperature change of the 0.40 mol of helium was the same size in both (451 K). **Is the heat needed per kelvin the same at constant pressure and at constant volume?**

Predict first

Which process makes the gas do work as well as warm up? Is the ratio of the two heats a simple number?

Poll

In the three-process cycle of Contact 2, the isobaric heating $A \rightarrow B$ took 3750 J and the isochoric cooling $B \rightarrow C$ released 2250 J. The temperature change of the 0.40 mol of helium was the same size in both (451 K). **Is the heat needed per kelvin the same at constant pressure and at constant volume?**

Predict first

Which process makes the gas do work as well as warm up? Is the ratio of the two heats a simple number?

The ratio is $3750/2250 = 5/3$. At constant pressure the heat must supply both the rise in U and the work of expansion. The heat per mole per kelvin: $\frac{3750}{(0.40)(451)} = 20.8$ against $\frac{2250}{(0.40)(451)} = 12.5 \text{ J}/(\text{mol K})$. These are the **molar heat capacities** C_p and C_V .

Rule: molar heat capacities

$$Q_V = nC_V\Delta T, \quad Q_p = nC_p\Delta T, \quad \Delta U = nC_V\Delta T$$

$$C_p = C_V + R, \quad \gamma = \frac{C_p}{C_V}$$

Q_V : constant volume. Q_p : constant pressure. ΔU : **every** process of an ideal gas.

Why $C_p = C_V + R$: at constant pressure,
 $Q = \Delta U + W$ gives $nC_p\Delta T = nC_V\Delta T + nR\Delta T$,
since $W = p\Delta V = nR\Delta T$.

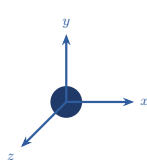
Common error

$\Delta U = nC_V\Delta T$ holds for every process, not only at constant volume. $Q = nC_p\Delta T$ holds only at constant pressure.

Defining heat capacities and degrees of freedom

Quantity	What it is	What it measures	SI unit	Representation
Molar heat capacities C_V , C_p	the energy per mole per kelvin at constant volume or constant pressure	how much energy a gas needs per kelvin in each case	J/(mol K)	a table value; $\frac{f}{2}R$
Degrees of freedom f	the independent ways a molecule can store energy	how many energy terms share the heat	—	3 monatomic, 5 diatomic
Adiabatic exponent γ	the ratio C_p/C_V	how steeply the adiabat falls	none	5/3 or 7/5
Interpret: the heat capacity answers <i>how much energy does this gas need per kelvin, and where does it go?</i>				

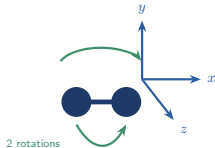
Equipartition and degrees of freedom



monatomic (He, Ar)

3 translations: $f = 3$

$$U = \frac{3}{2}nRT, C_V = \frac{3}{2}R$$



diatomic (N₂, O₂)

3 translations + 2 rotations: $f = 5$

$$U = \frac{5}{2}nRT, C_V = \frac{5}{2}R$$

Rule: equipartition

Each quadratic energy term has an average of $\frac{1}{2}kT$. With f degrees of freedom:

$$U = \frac{f}{2}nRT, \quad C_V = \frac{f}{2}R, \quad \gamma = \frac{f+2}{f}$$

At room temperature: monatomic $f = 3$; diatomic $f = 5$ (three translations and two rotations; vibration is not excited). So monatomic $C_V = \frac{3}{2}R = 12.5$, $\gamma = \frac{5}{3}$; diatomic $C_V = \frac{5}{2}R = 20.8$, $\gamma = \frac{7}{5}$ (in J/(mol K)).

Measured molar heat capacities

Type	Gas	C_V	C_p	γ
monatomic	He	12.47	20.78	1.67
	Ar	12.47	20.78	1.67
diatomic	H ₂	20.42	28.74	1.41
	N ₂	20.76	29.07	1.40
	O ₂	20.85	29.17	1.40
polyatomic	CO ₂	28.46	36.94	1.30

Young and Freedman, Table 19.1 (J/(mol K)).
 $C_p - C_V \approx R = 8.31$ for all.

Physical meaning

Measured values agree with $\frac{3}{2}R$ and $\frac{5}{2}R$. Polyatomic molecules have extra terms and a larger C_V .

Model validity

C_V is taken as constant: at low T rotations freeze out, at high T vibrations switch on (supplement).

EQUATION – molar heat capacities

$$Q_V = nC_V\Delta T, \quad Q_p = nC_p\Delta T$$
$$C_p = C_V + R$$

Meaning. energy per mole per kelvin;
constant pressure also pays for the work.

Variables. n ; ΔT ; C_V , C_p .

Units. J.

Interpretation. $Q_p - Q_V = nR\Delta T = W$.

Example. N_2 : 4.36 kJ against 3.12 kJ.

EQUATION – equipartition

$$U = \frac{f}{2}nRT, \quad C_V = \frac{f}{2}R$$
$$\gamma = \frac{f+2}{f}$$

Meaning. each degree of freedom stores $\frac{1}{2}kT$
per molecule.

Variables. f degrees of freedom.

Units. J; J/(mol K).

Interpretation. monatomic $f = 3$, diatomic
 $f = 5$.

Example. helium: $C_V = 12.5$, $\gamma = 1.67$.

TOOLKIT 5 – Ideal-gas energy and heat capacities

- 1. IDENTIFY THE GAS:** count the degrees of freedom: $f = 3$ (monatomic), $f = 5$ (diatomic at room temperature); $C_V = \frac{f}{2}R$, $C_p = C_V + R$, $\gamma = C_p/C_V$.
 - 2. ENERGY:** $\Delta U = nC_V\Delta T$ for every process.
 - 3. HEAT:** $Q = nC_V\Delta T$ at constant V ; $Q = nC_p\Delta T$ at constant p . In any other process, $Q = \Delta U + W$.
 - 4. WORK:** from Toolkit 2, or $W = nR\Delta T$ at constant pressure.
- WATCH OUT:** $Q = nC_V\Delta T$ outside a constant-volume process; $f = 5$ for a monatomic gas; °C differences are fine for ΔT but not for T .
- CHECK:** $Q_p > Q_V$ for the same ΔT , and $Q_p - Q_V = W = nR\Delta T$.

Example 4: nitrogen at constant volume and at constant pressure

Question

1.5 mol of nitrogen gas (N_2 , treat as an ideal diatomic gas) is warmed from 300 K to 400 K. Find Q , W and ΔU (a) at constant volume and (b) at constant pressure.

Predict first

In which case is more heat needed? By how much more, in terms of the work? Is ΔU the same in both cases?

Example 4: nitrogen at constant volume and at constant pressure

Question

1.5 mol of nitrogen gas (N_2 , treat as an ideal diatomic gas) is warmed from 300 K to 400 K. Find Q , W and ΔU (a) at constant volume and (b) at constant pressure.

Predict first

In which case is more heat needed? By how much more, in terms of the work? Is ΔU the same in both cases?

$$f = 5: C_V = \frac{5}{2}R = 20.79, C_p = \frac{7}{2}R = 29.10 \text{ J/(mol K)}; \Delta T = 100 \text{ K.}$$

$$(a) W = 0; \Delta U = Q = nC_V\Delta T = (1.5)(20.79)(100) = 3.12 \times 10^3 \text{ J.}$$

$$(b) \Delta U = 3.12 \times 10^3 \text{ J (same } \Delta T); Q = nC_p\Delta T = (1.5)(29.10)(100) = 4.36 \times 10^3 \text{ J;}$$
$$W = nR\Delta T = 1.25 \times 10^3 \text{ J.}$$

$$\textbf{Evaluate: } Q_p - Q_V = 1.25 \text{ kJ} = W, \text{ and } Q_p/Q_V = 1.40 = \gamma. \checkmark$$

The adiabatic process

- **Adiabatic** means $Q = 0$: no heat crosses the boundary. Either the walls are insulating, or the process is so fast that there is no time for heat to flow (but still slow enough for the gas to be in equilibrium at every step).
- The first law becomes $\Delta U = -W$: the work comes out of the internal energy.
- Expansion: $W > 0$ so U falls and the gas **cools**. Compression: the gas **warms**.
- Examples: a bicycle pump, a spray can, the compression in a diesel engine, and sound waves.

Sketch, then predict

From the same initial state, an ideal gas expands to twice its volume, once isothermally and once adiabatically. Sketch both on a pV diagram. Which ends at the lower pressure?

Deriving the adiabatic relation

1. **First law, $Q = 0$:** $dU = -p dV$, and for an ideal gas $dU = nC_V dT$. So $nC_V dT = -p dV$.
2. **Eliminate p** with $p = nRT/V$: $C_V dT = -\frac{RT}{V} dV$, that is $\frac{dT}{T} = -\frac{R}{C_V} \frac{dV}{V}$.
3. **Integrate;** with $R/C_V = \gamma - 1$ (from $C_p = C_V + R$): $\ln T + (\gamma - 1) \ln V = \text{constant}$.

Rule: adiabatic process of an ideal gas

$$TV^{\gamma-1} = \text{const}, \quad pV^{\gamma} = \text{const}, \quad W = nC_V(T_1 - T_2) = \frac{p_1 V_1 - p_2 V_2}{\gamma - 1}$$

Valid for an ideal gas with constant γ , in a quasi-static process with $Q = 0$. (Use $T = pV/nR$ to change between the first two.)

EQUATION – adiabatic process, ideal gas

$$pV^\gamma = \text{const}, \quad TV^{\gamma-1} = \text{const}$$

$$W = nC_V(T_1 - T_2)$$

Meaning. how p , V and T are linked when no heat crosses the boundary.

Variables. $\gamma = C_p/C_V$; T kelvin.

Units. J for W .

Interpretation. steeper than an isotherm; expansion cools, compression warms.

Example. diesel: 889 K, 48.5 atm.

Predict first

An ideal gas expands adiabatically. Predict whether T rises or falls, and whether the final pressure is above or below the isothermal value.

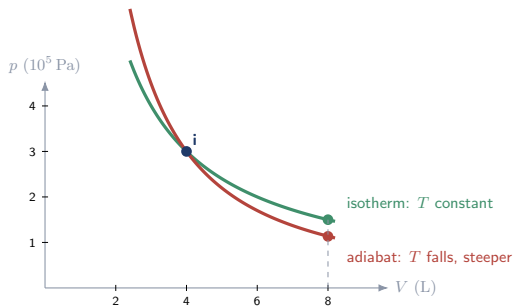
TOOLKIT 6 – Adiabatic scaling

1. **VERIFY:** $Q = 0$ (insulated or fast), an ideal gas, a smooth quasi-static change.
2. **GET** γ from the degrees of freedom (monatomic $\frac{5}{3}$, diatomic $\frac{7}{5}$) or from the table.
3. **SCALE:** $\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$, $\frac{p_2}{p_1} = \left(\frac{V_1}{V_2}\right)^{\gamma}$; T in kelvin.
4. **WORK:** $W = nC_V(T_1 - T_2)$, or $W = -\Delta U$.
5. **SKETCH:** the adiabat is steeper than the isotherm through the same point.

WATCH OUT: using $pV = \text{const}$ (isothermal); using $^{\circ}\text{C}$.

CHECK: expansion: $T_2 < T_1$ and $W > 0$; compression: $T_2 > T_1$ and $W < 0$.

Isotherm against adiabat



A diatomic gas ($\gamma = 1.40$) expands from i ($4.0 \text{ L}, 3.0 \times 10^5 \text{ Pa}$) to 8.0 L .

	isotherm	adiabat
p_f (10^5 Pa)	1.50	1.14
T_f/T_i	1	0.758
W (J)	832	726

The adiabat ends lower: the gas cools as it does work, so its pressure falls faster. It also does less work, because the heat that the isothermal gas takes in is missing: $W_{\text{ad}} = -\Delta U$ comes out of U .

Check: $p_f = 3.0 \times 10^5 \times 2^{-1.4}$; $W_{\text{iso}} = 1200 \ln 2$.

In Lecture 10 the air in a pump was compressed from 0.60 L to 0.12 L (a factor of 5) from 1.00 atm and 15°C. Compare three cases:

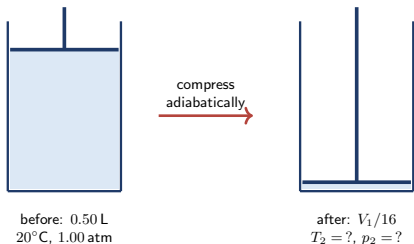
Limit	p_2 (atm)	T_2 (°C)
isothermal	5.00	15
Lecture 10 (as stated)	5.78	60
adiabatic ($\gamma = 1.40$)	9.52	275

Physical meaning

The real pump lies between the two limits: some heat escapes through the walls during the stroke, so the process is neither isothermal nor adiabatic. The adiabatic limit gives the highest temperature and pressure for a given compression.

Adiabatic values: $p_2 = 1.00 \times 5^{1.4}$ and $T_2 = 288.15 \times 5^{0.4} = 548.5$ K.

Example 5: compression in a diesel engine



Air ($\gamma = 1.40$) at 20°C and 1.00 atm fills a cylinder of volume 0.50 L . It is compressed adiabatically to $\frac{1}{16}$ of its volume. Find (a) the final temperature, (b) the final pressure, (c) the work done by the gas.

Predict first

Will the final temperature be above or below 100°C ? Will the pressure rise by more or less than the factor 16?

Set up: $T_1 = 293.15 \text{ K}$, $p_1 = 1.013 \times 10^5 \text{ Pa}$, $V_1/V_2 = 16$;

$$n = \frac{p_1 V_1}{RT_1} = \frac{(1.013 \times 10^5)(0.50 \times 10^{-3})}{(8.314)(293.15)} = 0.0208 \text{ mol}; \text{ diatomic, } C_V = 20.79 \text{ J/(mol K)}.$$

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$$\text{(a)} \quad T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{\gamma-1} = 293.15 \times 16^{0.40} = 889 \text{ K} = \mathbf{616^\circ\text{C}}.$$

$$\text{(b)} \quad p_2 = p_1 \left(\frac{V_1}{V_2} \right)^{\gamma} = 1.00 \text{ atm} \times 16^{1.40} = \mathbf{48.5 \text{ atm}} = 4.91 \times 10^6 \text{ Pa}.$$

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$$\text{(c)} \quad W = nC_V(T_1 - T_2) = (0.0208)(20.79)(293.15 - 888.7) = \mathbf{-257 \text{ J}}.$$

Evaluate: T_2 far above 100°C and p rises by 48.5, more than 16, because the temperature rises as well. $W < 0$ (compression). Check:

$$T_2 = p_2 V_2 / (nR) = (4.91 \times 10^6)(3.125 \times 10^{-5}) / (0.0208 \times 8.314) = 889 \text{ K}, \text{ and}$$
$$(p_1 V_1 - p_2 V_2) / (\gamma - 1) = -257 \text{ J}. \quad \checkmark$$

Your turn: adiabatic expansion of argon

Independent problem

0.30 mol of argon at 400 K occupies 6.0 L. It expands adiabatically to 12.0 L. Find the initial pressure, the final temperature and pressure, and the work done by the gas.

Predict first

Sketch the isotherm and the adiabat. Is T_2 above or below 400 K? Is the work more or less than for an isothermal expansion?

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Predict first

Sketch the isotherm and the adiabat. Is T_2 above or below 400 K? Is the work more or less than for an isothermal expansion?

Argon is monatomic: $\gamma = \frac{5}{3}$, $C_V = \frac{3}{2}R$. $p_1 = \frac{nRT_1}{V_1} = \frac{(0.30)(8.314)(400)}{6.0 \times 10^{-3}} = 1.66 \times 10^5 \text{ Pa}.$

$$T_2 = 400 \left(\frac{1}{2}\right)^{2/3} = 252 \text{ K}; \quad p_2 = \frac{nRT_2}{V_2} = 5.24 \times 10^4 \text{ Pa} \quad (\text{check: } p_1 \left(\frac{1}{2}\right)^{5/3}).$$

$$W = nC_V(T_1 - T_2) = (0.30)(12.47)(148) = 554 \text{ J} \quad (\text{isothermal expansion: } nRT \ln 2 = 692 \text{ J}).$$

Evaluate: cooler and a smaller work than the isothermal case, as predicted. ✓

Concept check: compression to half the volume

Choose, then explain

An ideal gas is compressed to half its volume from the same initial state, once isothermally and once adiabatically. Which compression requires more work to be done *on* the gas?

A. the isothermal one B. the adiabatic one C. the same for both

Predict first

Sketch both paths on a pV diagram. Which path is higher, and what does the area under it represent?

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Predict first

Sketch both paths on a pV diagram. Which path is higher, and what does the area under it represent?

B. Compression moves left. The adiabat is steeper, so its pressure is higher at every volume and the area under it is larger. For $\gamma = 1.40$: $|W_{\text{ad}}| = 2.5 nRT_1(2^{0.4} - 1) = 0.80 nRT_1$ against $|W_{\text{iso}}| = nRT_1 \ln 2 = 0.69 nRT_1$.

TOOLKIT 7 – Cycle analysis

1. **STATES:** find p , V and T at every corner, with $pV = nRT$ and, for an adiabatic leg, $pV^\gamma = \text{const.}$
 2. **LEGS:** name the process on each leg (Toolkit 4).
 3. **LEDGER:** for each leg find W (Toolkit 2), $\Delta U = nC_V\Delta T$ (Toolkit 5) and $Q = \Delta U + W$ (Toolkit 3).
 4. **CLOSE THE CYCLE:** $\sum \Delta U = 0$ and $\sum Q = \sum W =$ the enclosed area.
 5. **SPLIT:** add the positive Q (heat absorbed) and the negative Q (heat rejected) separately.
- WATCH OUT:** a corner state found from the wrong leg; the sign of W on a leg that goes to smaller V .
- CHECK:** the cycle closes; heat absorbed minus heat rejected equals the net work.

Lecture 12: engines

The cycle of Contact 3 delivers 331 J of work from 2000 J of heat. The efficiency of an engine and its limit follow from the second law.

Refrigerators

A counter-clockwise cycle moves heat from cold to hot with work: a refrigerator or heat pump.

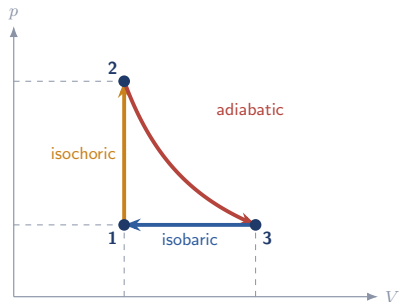
Atmosphere and engines

The adiabatic cooling of rising air sets weather and the lapse rate; adiabatic compression ignites fuel in a diesel engine.

Physical meaning

$\Delta U = Q - W$ is the conservation of energy of Lectures 4 and 5, with heat included.

Integrated problem: a three-leg cycle



Application problem

0.20 mol of nitrogen (diatomic, ideal) starts at state 1: 1.00×10^5 Pa, 4.0 L. The gas is heated at constant volume until its pressure is 3.00×10^5 Pa (state 2), then expands adiabatically until the pressure is again 1.00×10^5 Pa (state 3), then is compressed at constant pressure back to state 1.

- Find p , V and T at the three states.
- Find Q , W and ΔU for each leg.
- Find the net work and the net heat; compare.
- Which legs absorb heat?

Predict first

Sketch the cycle. Does it run clockwise? Predict the sign of W on each leg, and of the net work.

Integrated problem: the states

Identify, set up: an ideal diatomic gas, a closed cycle: Toolkits 4, 5, 6 and 7. $\gamma = 1.40$, $C_V = 20.79$, $C_p = 29.10$ J/(mol K); $nR = 1.663$ J/K.

State	p (10^5 Pa)	V (L)	T (K)	How found
1	1.00	4.0	240.6	given; $T = pV/nR$
2	3.00	4.0	721.7	isochoric: $T_2 = 3T_1$
3	1.00	8.77	527.3	adiabat: $V_3 = V_2(p_2/p_3)^{1/\gamma} = 4.0 \times 3^{0.714}$

Integrated problem: the states

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Check state 3: $T_3 = p_3 V_3 / nR = (1.00 \times 10^5)(8.77 \times 10^{-3}) / 1.663 = 527 \text{ K}$; also $T_3 = T_2(p_3/p_2)^{(\gamma-1)/\gamma} = 721.7 \times 3^{-0.2857} = 527 \text{ K}$. ✓

Integrated problem: the ledger

Leg	Process	Q (J)	W (J)	ΔU (J)
1→2	isochoric	2000	0	2000
2→3	adiabatic	0	808	-808
3→1	isobaric	-1669	-477	-1192
cycle	total	331	331	0

12: $Q = nC_V(T_2 - T_1) = (0.20)(20.79)(481.1) = 2000$. 23:

$W = nC_V(T_2 - T_3) = (0.20)(20.79)(194.4) = 808$. 31:

$W = p\Delta V = (1.00 \times 10^5)(4.0 - 8.77) \times 10^{-3} = -477$;

$Q = nC_p(T_1 - T_3) = (0.20)(29.10)(-286.7) = -1669$.

Integrated problem: the ledger

Leg	Process	Q (J)	W (J)	ΔU (J)
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$Q = nC_p(T_1 - T_3) = (0.20)(29.10)(-286.7) = -1669$.

Evaluate: (a), (b) above; (c) $W_{\text{net}} = Q_{\text{net}} = 331$ J, $\sum \Delta U = 0$: the cycle closes. (d) Only 1→2 absorbs heat (2000 J); 3→1 rejects 1669 J. Clockwise, so $W_{\text{net}} > 0$, as predicted. Of the 2000 J taken in, 331 J (16.6%) becomes net work: this ratio is the *efficiency* of an engine, the subject of Lecture 12. ✓

Extension (unassessed): a rising parcel of air cools

Air rising in the atmosphere expands as the pressure around it falls. Heat conducts very slowly through air, so a rising parcel is nearly adiabatic.

A parcel at 20°C rises to a height where the pressure is 0.80 atm (about 2 km above sea level).

With $\gamma = 1.40$:

$$T_2 = T_1 \left(\frac{p_2}{p_1} \right)^{(\gamma-1)/\gamma} = 293.15 \times 0.80^{0.2857} = 275.0 \text{ K} = 1.9^\circ\text{C}.$$

Not assessed. The altitude is read from a standard atmosphere; the relation between pressure and height is taught with fluid statics.

Lapse rate

The parcel cools by about 17 K over about 2 km, roughly 9 K/km. This is close to the measured dry adiabatic lapse rate of about 10 K/km: expansion alone, with no heat lost, cools the air.

Common mistakes and the correct habit

Mistake

Using $Q = nC_V\Delta T$ outside constant volume.

Using $pV = \text{const}$ for an adiabat.

Taking $f = 5$ for a monatomic gas.

Using $^{\circ}\text{C}$ in $TV^{\gamma-1}$.

Finding a corner state from the wrong leg.

Instead

Use $\Delta U = nC_V\Delta T$; $Q = \Delta U + W$.

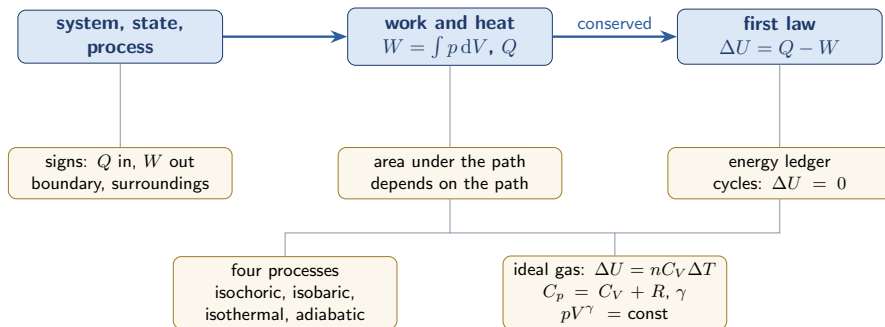
$pV^{\gamma} = \text{const}$.

$f = 3$ monatomic, 5 diatomic.

Kelvin.

Name the process on each leg and follow its rule.

Lecture 11 concept map



MASTER PROCEDURE – THE FIRST LAW

1. **IDENTIFY** The system, the substance (ideal gas? monatomic or diatomic?), and what is fixed.
 2. **SKETCH** The path on a pV diagram, with arrows; label every state.
 3. **SIGNS** Q in or out? W by or on the system? Units in joules and kelvin.
 4. **PREDICT** The sign of W , Q , ΔU and ΔT for each leg.
 5. **SELECT THE MODEL** Area or integral, first law, process rule, C_V and C_p , adiabat.
 6. **EXECUTE** Fill in the ledger, one row for each leg, equation first.
 7. **EVALUATE** Units, signs, ΔU the same on every path, the cycle closes, limiting cases.
- SKETCH $\rightarrow$ PREDICT $\rightarrow$ CALCULATE $\rightarrow$ CHECK.

Inside the course framework

IDENTIFY $\leftrightarrow$ steps 1–2
SET UP $\leftrightarrow$ steps 3–5
EXECUTE $\leftrightarrow$ step 6
EVALUATE $\leftrightarrow$ step 7

Physical meaning

Next lecture: the cycle of Contact 3 delivers net work 331 J from 2000 J of heat. Why can the ratio never reach one? That is the second law, heat engines and entropy.

THE SEVEN TOOLKITS

- | | | |
|---|---|---|
| 1 | System and sign convention | $Q > 0$ in; $W > 0$ by the system; joules |
| 2 | Work from a pV diagram | area; sign by direction; cycles |
| 3 | The energy ledger | $\Delta U = Q - W$ row by row; state function |
| 4 | Recognising the process | isochoric, isobaric, isothermal, adiabatic |
| 5 | Ideal-gas energy and heat capacities | $\Delta U = nC_V\Delta T$; $C_p = C_V + R$; f |
| 6 | Adiabatic scaling | $TV^\gamma = \text{constant}$, $pV^\gamma = \text{constant}$; steeper than isotherm |
| 7 | Cycle analysis | states, ledger per leg, closure |

The Master Procedure sits above these. Earlier toolkits (units, energy, signs, the ideal-gas equation, the ratio method, $Q = mc\Delta T$) are reused, not renamed.

Key ideas

- Q and W depend on the path; ΔU does not.
- $W > 0$ on expansion; $Q > 0$ for heat in.
- For a cycle, $\Delta U = 0$ and $Q_{\text{net}} = W_{\text{net}}$.
- For an ideal gas, $\Delta U = nC_V\Delta T$ in every process.
- Isothermal: T constant; adiabatic: $Q = 0$.
- The adiabat is steeper than the isotherm.

Where to look

Definitions: the vocabulary and equation-anatomy slides of each contact.

Common mistakes: the last slide of each contact.

Connections: the Connect boxes and slides.

Summary: core equations

Idea	Equation	Conditions and units
Work by a gas	$W = \int_{V_1}^{V_2} p dV$	area; $W = p\Delta V$ at constant p
Isothermal work	$W = nRT \ln(V_2/V_1)$	ideal gas; J
First law	$\Delta U = Q - W$	$Q > 0$ in, $W > 0$ by the system
Cycle	$Q_{\text{net}} = W_{\text{net}}$	$\Delta U = 0$
Ideal-gas energy	$U = \frac{f}{2}nRT, \quad \Delta U = nC_V\Delta T$	any process
Heat capacities	$Q_V = nC_V\Delta T, \quad Q_p = nC_p\Delta T, \quad C_p = C_V + R$	J/(mol K)
Adiabatic	$pV^\gamma = \text{const}, \quad TV^{\gamma-1} = \text{const}$	$Q = 0$, ideal gas
Adiabatic work	$W = nC_V(T_1 - T_2)$	J

Checklist: can I ...

- ☐ name the system and classify it as open, closed or isolated?
- ☐ state the sign convention for Q and W ?
- ☐ derive and use $W = \int p dV$?
- ☐ find the work from a pV diagram, with its sign?
- ☐ explain why W depends on the path?
- ☐ describe internal energy as a state function?
- ☐ apply $\Delta U = Q - W$ and build a ledger?
- ☐ treat a cycle (net work equals net heat)?
- ☐ recognise the four processes and give W , Q , ΔU ?
- ☐ use the internal energy of an ideal gas?
- ☐ use C_V , $C_p = C_V + R$ and γ ?
- ☐ relate C_V to the degrees of freedom?
- ☐ use $pV^\gamma = \text{const}$ and $TV^{\gamma-1} = \text{const}$?
- ☐ compare an adiabat with an isotherm?
- ☐ analyse a cycle for net work and net heat?
- ☐ select the model in an unfamiliar problem?

After this week

Tutorial sheet: signs, work from areas, ledger and cycle problems, the four processes, boiling water, heat capacities, adiabatic problems and a combined problem where the method is not given.

Supplementary notes: Joule's experiment, the isothermal and adiabatic derivations, C_V against temperature, polyatomic gases, the sign convention $\Delta U = Q + W$, enthalpy, and a preview of Lecture 12.